NumPy Practice: Real-World Exercises

This document presents a set of NumPy practice exercises designed to enhance understanding and application of the NumPy library in real-world scenarios.

Exercise 1: Student Marks Analysis

Problem Statement: Marks of 5 students are stored in a NumPy array:

```
marks = [45, 78, 62, 90, 55]
```

Tasks:

- Find the average marks
- Find the highest and lowest marks
- Count how many students scored above average

Exercise 2: Temperature Data (5 Days)

Problem Statement: Temperature of a city for 5 days (in Celsius) is stored in a NumPy array:

```
temp = [30, 32, 35, 38, 33]
```

Tasks:

- Convert temperature to Fahrenheit
- Find the hottest and coldest day
- Count how many days temperature was above 34°C

Exercise 3: Sales Report (5 Days, 3 Shops)

Problem Statement: Daily sales of 3 shops for 5 days are stored in a 2D NumPy array:

```
sales = [
    [200, 250, 300, 280, 260],
    [180, 220, 270, 260, 240],
    [210, 260, 310, 290, 270]
]
```

Tasks:

- Find total sales of each shop
- Find the shop with highest total sales
- Find the average sales per day

Exercise 4: Product Price Update

Problem Statement: Prices of 5 products are stored in a NumPy array:

```
prices = [150, 450, 600, 800, 300]
```

Tasks:

- Add 10% tax to all products
- Give 20% discount to products priced above 500 (after tax)
- Print the final price list

Exercise 5: Attendance Tracker (5 Students, 5 Days)

Problem Statement: Attendance of 5 students for 5 days is stored in a NumPy array (1 = Present, 0 = Absent):

```
attendance = [
    [1, 1, 1, 0, 1],
    [1, 0, 1, 1, 1],
    [0, 1, 1, 0, 1],
    [1, 1, 1, 1, 1],
    [1, 0, 0, 1, 1]
]
```

Tasks:

- Find total attendance of each student
- Find students with attendance below 4 days
- Find the most regular student